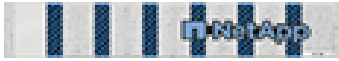
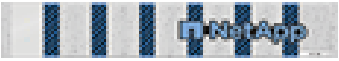
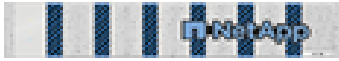
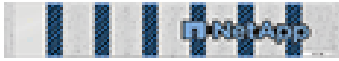
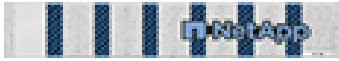
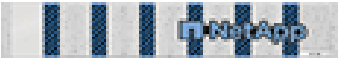
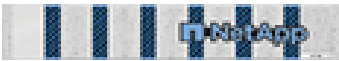


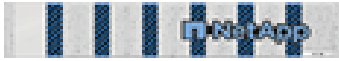
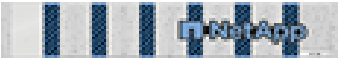
Specifications	AFX 1K Cluster	AFX 2K Cluster
		
	ONTAP 9.19.1	ONTAP 9.19.1
Storage Limits		
Max Number of Drives (Total)	600	600
Internal Drives count	N/A	N/A
Max number of NVMe SSD Drives	Max: 600 per Cluster;	Max: 600 per Cluster;
Max number of NX224 shelves	25 external	25 external
Capacity Limits		
Marketing Maximum Raw Capacity (GB)		
Cluster	36,840,000	36,840,000
Core Cluster Limits		
Maximum number of storage virtual machines (SVMs) - NAS		
Cluster	1,024	1,024
Maximum number of LIFs - NAS		
Node	1,024	1,024
Cluster	4,096	4,096
Maximum number of connections - NAS/S3		
Node	100,000	100,000
Maximum Number of VLANs		
Node	512	512
Maximum number of flexible volumes - NAS		
Node	2,500	2,500
Cluster	30,000	30,000
Maximum number of lock manager locked objects (CIFS and NFS combined)		
Node	3,000,000	3,000,000
Cluster	12,288,000	12,288,000
Maximum number of Intercluster LIFs		
Node	8	8
Minimum Root Volume (GiB)		
Node	150	150
Maximum IPsec Policies		
SVM	256	256
Cluster	1,500	1,500
Maximum Number of IPspaces		
Cluster	512	512
Maximum Number of Auditing-Enabled SVMs		
Cluster	400	400
System Maximums and Limits		
Max Nodes per Cluster (NAS / SAN)	32 / Not Supported	Not Supported / Not Supported
Max FabricPool Size	291038 TiB	291038 TiB
Max FlexGroup Data Constituent Size	300 TiB	300 TiB
Max Infinite Volume Data Constituent Size	Not Supported	Not Supported
Min Root Volume Size	150 GiB	150 GiB
NetApp Volume Encryption	NVE	NVE
Processor		
Processor Model	64-bit 52-Core 1.7GHz	64-bit 48-Core 2.3GHz
Processor Architecture	64 bit	64 bit
Processor Speed	1.70 Ghz	2.30 Ghz
Processor Count (Per Node)	2	2

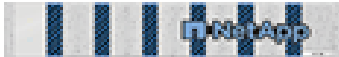
Specifications	AFX 1K Cluster	AFX 2K Cluster
		
	ONTAP 9.19.1	ONTAP 9.19.1
Processor Count (Per Config)	2	2
Processor Cores (Per CPU)	52	48
Processor Cores (Per Node)	104	96
Processor Cores (Per Config)	104	96
Memory		
NVRAM (Per Node)	64 GB	128 GB
NVRAM (Per Config)	64 GB	128 GB
RAM (Per Node)	1024 GB	1024 GB
RAM (Per Config)	1024 GB	1024 GB
Onboard Ports		
Ethernet Ports	-	-
UTA2 Ports	-	-
Expansion Slots	9 x IO Module (per System)	9 x IO Module (per System)
SAS Ports	-	-
Physical Characteristics		
Rack Units	2	2
Chassis Height	3.5" (8.89 cm)	3.5" (8.89 cm)
Chassis Width with Mounting Flanges	19.02" (48.3 cm)	19.02" (48.3 cm)
Chassis Width without Mounting Flanges	17.68" (44.9 cm)	17.68" (44.9 cm)
Chassis Depth with Cable Mgmt	35.24" (89.5 cm)	35.24" (89.5 cm)
Chassis Depth without Cable Mgmt	30.04" (76.3 cm)	30.04" (76.3 cm)
Chassis Weight		
System Clearance Dimensions		
Front Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)
Front Clearance (Maintenance)	31.99" (81.2 cm)	31.99" (81.2 cm)
Rear Clearance (Cooling)	6.03" (15.3 cm)	6.03" (15.3 cm)
Rear Clearance (Maintenance)	11.03" (28 cm)	11.03" (28 cm)
Environmental Requirements		
Operating Temperature Range	50 to 95 deg F 10 to 35 deg C	50 to 95 deg F 10 to 35 deg C
Storage Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Transit Temperature Range	-40 to 158 deg F -40 to 70 deg C	-40 to 158 deg F -40 to 70 deg C
Operating Relative Humidity	8 to 80 %	8 to 80 %
Storage Relative Humidity	10 to 95 %	10 to 95 %
Transit Relative Humidity	10 to 95 %	10 to 95 %
Operating Altitude Range	Up to 10000.0 ft Up to 3048.0 m	Up to 10000.0 ft Up to 3048.0 m
Storage Altitude Range	Up to 40000.0 ft Up to 12192.0 m	Up to 40000.0 ft Up to 12192.0 m
Transit Altitude Range	Up to 39989.8 ft Up to 12192 m	Up to 39989.8 ft Up to 12192 m
Acoustic Noise - Sound Power	8.5 Bels	8.5 Bels
Acoustic Noise - Sound Pressure	67.7 dBA	67.7 dBA
Input Power Voltage	200 to 240;	200 to 240;
Storage OS Requirements (for selected major version)		
Recommended Version	9.19.1, 9.19.1P2	9.19.1, 9.19.1P2
Minimum OS	9.19.1RC0	9.19.1
Maximum OS	9.19.1P2	9.19.1P2

Specifications	AFX 1K Cluster	AFX 2K Cluster
		
	ONTAP 9.19.1	ONTAP 9.19.1
Software and Firmware * indicates that firmware is bundled with Storage OS Version		
BIOS	20.8*	20.8*
BMC	18.6*	18.6*
Product Standards Compliance		
Certifications EMC/EMI	AMCA, FCC, ICES, KC, Morocco, VCCI	AMCA, FCC, ICES, KC, Morocco, VCCI
Certifications safety	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS	BIS, CB, CSA, G_K_U-SoR, IRAM, NOM, NRCS, SONCAP, TBS
Certifications Safety/EMC/EMI	EAC, UKRSEPRO	EAC, UKRSEPRO
Certifications Safety/EMC/EMI/RoHS	BSMI, CE DoC, UKCA DoC	BSMI, CE DoC, UKCA DoC
Standards EMC/EMI	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835	BS-EN-55032, BS-EN55035, CISPR 32, EN55022, EN55024, EN55032, EN55035, EN61000-3-2, EN61000-3-3, FCC Part 15 Class A, ICES-003, KS C 9832, KS C 9835
Standards Safety	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)	ANSI/UL60950-1, ANSI/UL62368-1, BS-EN62368-1, CAN/CSA C22.2 No. 60950-1, CAN/CSA C22.2 No. 62368-1, CNS 15598-1, EN60825-1, EN62368-1, IEC 62368-1, IEC60950-1, IS 13252(part 1)
System Availability & Support		
Release Date	Oct 2025	Jun 2026
End of Availability (EOA)	-	-
End of Support (EOS)	-	-
NFS Cluster Limits		
Maximum Number of NFS over TLS Connections		
Node	10,000	10,000
Maximum number of Export Policies		
Cluster	24,000	24,000
Maximum number of Export Rules		
Cluster	200,000	200,000
Maximum NFSv4 Access Control Entries		
ACL (Access List)	1,024	1,024
Maximum number of client objects		
Node	100,000	100,000
Maximum Number of NFSv4 ByteLock Objects		
Node	53,248,000	53,248,000
Maximum Number of NFSv4 Client Objects		
Node	1,638,400	1,638,400
Maximum Number of NFSv4 Copy State Objects		
Node	16,384,000	16,384,000
Maximum Number of NFSv4 Deleg Objects		
Node	32,768,000	32,768,000
Maximum Number of NFSv4 Deleg State Objects		
Node	32,768,000	32,768,000
Maximum Number of NFSv4 Layout Objects		
Node	16,384,000	16,384,000
Maximum Number of NFSv4 Layout State Objects		
Node	16,384,000	16,384,000

Specifications	AFX 1K Cluster		AFX 2K Cluster	
				
	ONTAP 9.19.1		ONTAP 9.19.1	
Maximum Number of NFSv4 LayoutLock Objects				
Node	16,384,000		16,384,000	
Maximum Number of NFSv4 Lock State Objects				
Node	32,768,000		32,768,000	
Maximum Number of NFSv4 Open Objects				
Node	49,152,000		49,152,000	
Maximum Number of NFSv4 Open State Objects				
Node	49,152,000		49,152,000	
Maximum Number of NFSv4 Owner Objects				
Node	9,830,400		9,830,400	
Maximum Number of NFSv4 Sessions				
Node	1,638,400		1,638,400	
Maximum Number of NFSv4 String Objects				
Node	16,384,000		16,384,000	
WAFL Cluster Limits				
Maximum Size of a 64-bit Volume (TiB)				
Volume	300		300	
Maximum file size in a 64-bit Volume (TiB)				
File	128		128	
Maximum number of Volume Snapshot Copies				
Volume	1,023		1,023	
Node	1,023,000		1,023,000	
Maximum character length for Snapshot copy names				
Snapshot	255		255	
Maximum number of hard links				
File	100,000		100,000	
Maximum number of inodes/files				
Volume	2,000,000,000		2,000,000,000	
Maximum number of qtrees				
Volume	4,995		4,995	
Node	100,000		100,000	
Cluster	2,400,000		2,400,000	
Maximum number of qtrees with Extended Performance Monitoring				
Cluster	50,000		50,000	
Maximum number of concurrent DataMotion for Volumes (vol move) operations				
Node	16		16	
Maximum Depth of Clones				
Volume	1,000		1,000	
Maximum Number of First Origin Clones				
Volume	1,023		1,023	
Quality of Service Cluster Limits				
Maximum number of Policy Groups				
Cluster	12,000		12,000	
Maximum number of QoS user workloads				
Cluster	40,000		40,000	
CIFS Cluster Limits				

Specifications	AFX 1K Cluster		AFX 2K Cluster	
				
	ONTAP 9.19.1		ONTAP 9.19.1	
Maximum number of connected shares				
Node	1,000,000		1,000,000	
Maximum number of regular shares				
Cluster	300,000		300,000	
Maximum number of open files				
Connection	65,536		65,536	
Node	1,500,000		1,500,000	
Maximum number of local users				
Cluster	35,000		35,000	
Maximum number of local groups				
Cluster	320,000		320,000	
Maximum number of local group members				
Cluster	640,000		640,000	
Data Protection Cluster Limits				
Maximum number of NDMP sessions				
Node	36		36	
Maximum number of data protection (DP) mirrors and/or SnapVault relationships				
Volume	1		1	
Node	2,500		2,500	
Cluster	15,000		15,000	
Maximum number of concurrent Snap Mirror or SnapVault transfers				
Node	250		250	
Maximum fan-out from source for DP mirror				
Volume	20		20	
Maximum number of clusters that can be peered				
Cluster	1,023		1,023	
Maximum Number of constituent volumes in a SnapMirror relationship				
FlexGroup	200		200	
FlexGroup/Node	50		50	
Cluster	6,000		6,000	
Maximum number of SnapMirror Synchronous relationships				
Volume	1		1	
Node	400		400	
Maximum Number of Flexible Volumes in SVM DR				
SVM	300		300	
Maximum Number of storage virtual machine disaster recovery (SVM DR) relationship				
Cluster	128		128	
Maximum Number of Concurrent SVM Migration				
Cluster	1		1	
Maximum Number of Volumes with SVM Migration				
SVM	400		400	

Specifications	AFX 1K Cluster	AFX 2K Cluster
		
	ONTAP 9.19.1	ONTAP 9.19.1
Maximum Size of Cluster for SVM Migration		
Cluster	24	24
Maximum Number of FlexGroups and/or Consistency Groups supported in (SVM DR) relationship		
Cluster	32	32
Maximum Number of FlexGroups Constituents Volumes supported in (SVM DR) relationship		
FlexGroup	200	200
Maximum Number of Concurrent SnapDiff Sessions		
Node	64	64
FlexCache Cluster Limits		
Max caches from/of an origin volume (degree of fan-out)		
Origin	100	100
Max origin volumes		
Node	400	400
Max cache constituent volumes		
Node	800	800
Max cache volumes		
Node	400	400
Max origin constituent volumes		
Node	800	800
S3 Cluster Limits		
Maximum Number of Buckets		
FlexGroup Volume	1,000	1,000
Cluster	12,000	12,000
Maximum Number of Objects		
Bucket	2,000,000,000,000	2,000,000,000,000
Maximum Number of Users		
Object Store/StorageVM	4,000	4,000
Cluster	96,000	96,000
Maximum Size of the Bucket (PB)		
Vserver	60	60
Consistency Groups		
Maximum Number of CG Constituents Volumes supported in (SVM DR) relationship		
Consistency group	80	80
Maximum Number of Consistency Groups		
Cluster	15,000	15,000
Maximum Number of Consistency Groups and Parent Consistency Groups		
Cluster	15,000	15,000
Maximum Number of Parent Consistency Groups		
Cluster	15,000	15,000
Maximum Number of Child Consistency Groups in a Parent Consistency Group		

Specifications	AFX 1K Cluster	AFX 2K Cluster
		
	ONTAP 9.19.1	ONTAP 9.19.1
Parent consistency group	5	5
Maximum Number of Flexible Volumes in a Consistency Group		
Consistency group	80	80
Maximum Flexible Volumes across all the Child Consistency Groups under a Parent Consistency Group		
Parent consistency group	80	80
Maximum Number of Flexible Volumes in a Child Consistency Group		
Child consistency group	80	80
FlexGroup Rebalancing Limits		
Maximum Concurrent File Move at Destination Node		
Node	1,500	1,500
Maximum Number of Concurrent File Move Operations		
FlexGroup Constituent	1,500	1,500